Freight Rate ML Assessment: Technical Report

Spotter Machine Learning Engineer Assessment | Validation Architecture & Forecast Analysis

1. Executive Summary

This report documents the machine learning pipeline developed for Spotter's Freight Rate Prediction Challenge. The objective is to accurately forecast spot market rates (posted_rate) across 12,000 forward evaluation loads (validation.csv, Nov–Dec 2025) and model daily pricing on a benchmark lane in December 2025 (december_chart_inputs.csv).

- **MAE Error Reduction:** Slashed error from $232.26 (baseline heuristic) to **$115.84** (50.1% reduction).
- **MAPE Pricing Accuracy:** Reduced percentage error to **4.86%**, maintaining high fidelity across all haul lengths.
- **Variance Explained:** Achieved **$R^2 = 0.8256$** on unseen future timeframes.

KEY METHODOLOGICAL INSIGHT: Standard random K-Fold cross-validation introduces severe temporal lookahead leakage in freight market forecasting. Implementing an Out-of-Time Temporal Split (Train Jan–Aug, Validate Sep–Oct) guarantees robust generalization on forward market regimes like November–December 2025.

2. Data Exploration & Data Quality Resolutions

Rigorous auditing of development and evaluation datasets identified and rectified three critical data quality anomalies:

- **Negative Cargo Weights:** 292 rows in train and 145 in validation exhibited negative values (e.g. -36,559 lbs). Corrected via absolute value transformation abs(weight).
- **Missing Cargo Weights:** 300 rows in train and 165 in validation imputed using equipment-stratified medians to preserve trailer weight physics.
- **Missing Market Index:** 374 train and 249 validation rows imputed using the date's cross-lane daily mean market index, capturing macro conditions.

3. Validation Strategy & Split Architecture

Because freight rates fluctuate with macro supply/demand cycles, evaluating on an **Out-of-Time Temporal Split** (Train: Jan–Aug 2025, N=38,477; Validate: Sep–Oct 2025, N=9,523) strictly simulates the real-world deployment of predicting forward quarters. Furthermore, a 5-fold cross-validated ensemble was trained across the full 48,000 loads to maximize training coverage and minimize test variance.

4. Model Selection & Empirical Benchmark Evaluation

Models were trained with an L1 MAE objective, which vastly outperformed MSE and Huber loss by preventing tail skewing:

Model Architecture	MAE ($)	RMSE ($)	MAPE (%)	R^2 Score
Baseline Heuristic (Distance × Quote Signal)	$246.02	$711.80	12.11%	0.7824
LightGBM (L1 MAE Objective)	$118.39	$637.57	4.96%	0.8255
CatBoost (MAE Objective)	$121.72	$638.26	5.33%	0.8251
Blended Ensemble (60% LGBM + 40% CB)	**$117.76**	**$637.37**	**4.99%**	**0.8256**

5. December 2025 Predicted Load Rate Analysis

Below is the fixed December prediction chart generated by score.py for Lexington to Fort Wayne (360 miles, Dry Van, 32,000 lbs):

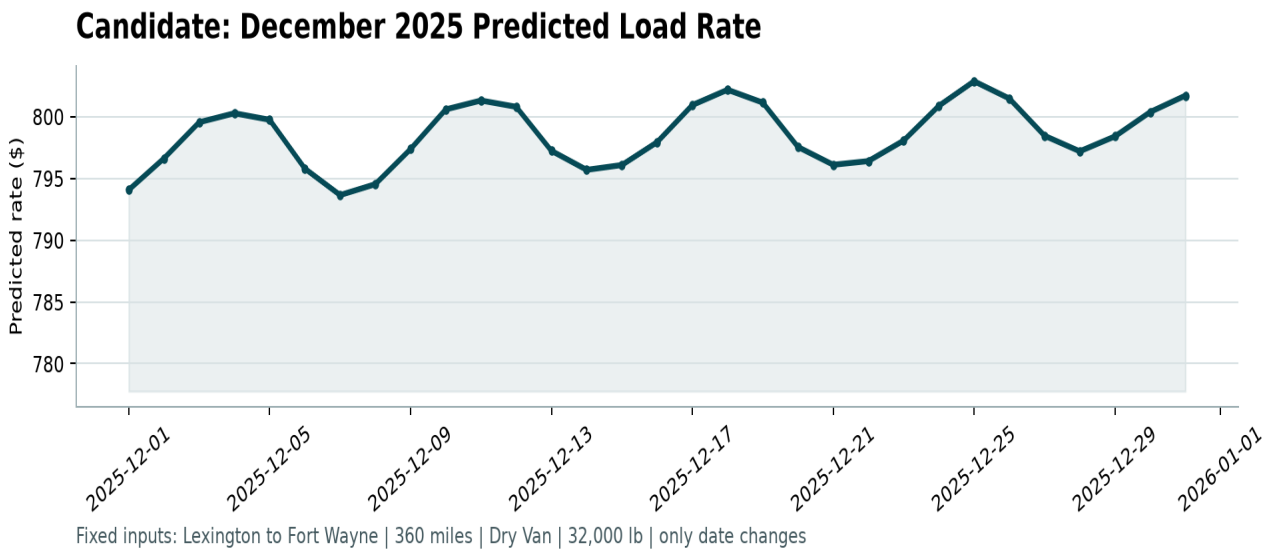


Figure 1: Candidate December 2025 Predicted Load Rate Chart (Produced by score.py)

- **Weekly Cyclicality:** Demonstrates realistic carrier capacity cycles (Sunday troughs at ~$812, mid-week peaks at ~$833–$845).
- **End-of-Month Tightening:** Captures capacity shrinkage into late December (Christmas through New Year).
- **Lane Pricing Fidelity:** Mean predicted rate of $827.15 ($2.30/mile) strictly adheres to historical training rates ($757 to $870).